Reproducibility Report: Guess and Coppock (2020)

Does Counter-Attitudinal Information Cause Backlash? Results from Three Large Survey Experiments

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This repository holds the actively maintained replication code for Guess and Coppock (2020), together with the reproducibility report that documents what the original archive did and did not do. It is part of a program applying the maintenance proposal in Peer, Orr and Coppock (2021, *PS: Political Science & Politics*, doi 10.1017/S1049096521000366) to a set of published archives.

Article	10.1017/S0007123418000327
Corrigendum	10.1017/S0007123419000024
Replication archive	10.7910/DVN/J7WNTM
Pre-analysis plans	osf.io/y9vdr, osf.io/5gdes

The data are not redistributed here. The deposit lives at Harvard Dataverse and that is the only copy this repository points at. `download_original.R` fetches it and checks every file against the MD5 checksums the archive itself publishes; `original_manifest.csv` records those checksums, the Dataverse file identifiers, and the UNF of each ingested data file, so the exact bytes this code was written against are pinned in version control even though the bytes themselves are not. Re-hosting a copy would create a second archive that can drift from the first, which is the problem this project exists to document rather than to add to.

Repository layout. `maintained/` is the maintained rewrite: one script per published table or figure, writing to `output/`, which is committed so a reader can compare a fresh run against it without downloading anything. `ground_truth/` ties every published number to the code that produces it. `original/` is created by the download script and is deliberately absent from the repository. This README is the reproducibility report, also available as a PDF in `report/`.

License. CC0 1.0 Universal, matching the terms of the deposit this repository maintains, so nothing in the chain is more restrictive than the archive itself. See `LICENSE`.

To reproduce.

```
Rscript download_original.R          # fetches the deposit, verifies 11 checksums
Rscript maintained/table_1_study1_conditions.R
# ... the remaining table scripts, in any order
Rscript maintained/figure_1_cate_estimates.R
Rscript maintained/figure_2_bart.R                # ~30 seconds, six BART fits
Rscript maintained/text_summary_stats.R           # last: reads both figures' estimates
Rscript maintained/text_bart_seed_sensitivity.R   # ~6 minutes, refits BART at ten seeds
```

Paths resolve through `here`, so the working directory does not matter. Required packages: tidyverse, estimatr, rsample, knitr, kableExtra, here. A successful run overwrites `maintained/output/`; `git diff` on that folder is the reproduction check, and rounded table output should come back byte-identical.

1 Paper Overview

Citation: Guess, A. and Coppock, A. (2020). Does counter-attitudinal information cause backlash? Results from three large survey experiments. *British Journal of Political Science*, 50(5), 1497–1515.

Research question: Does exposure to counter-attitudinal information cause backlash — attitude change in the direction opposite to the information — among individuals who oppose the position conveyed?

Design: Three pre-registered within-and-between-subjects survey experiments. Study 1 (N = 2,122): nationally-representative GfK sample, gun control issue, three-arm design (control, positive, negative information). Study 2 (N = 1,170): MTurk sample, minimum wage issue, thirteen-condition design crossing pro- and anti-minimum-wage videos. Study 3 (N = 683): MTurk sample, capital punishment issue, six-condition design varying combinations of pro- and anti-capital-punishment research reports. All studies measured pre-treatment baseline attitudes (T1) and post-treatment outcomes (T2). OLS with HC2 robust standard errors; covariate-adjusted and unadjusted estimates reported for proponents and opponents separately.

Main finding: Across all three studies, no evidence of backlash. Out of twenty-four treatment effect estimates (both CATE and CAPTE) across the three studies and two outcomes each, twenty-three are correctly signed. A formal binomial test rejects the null of equal probability of correct and incorrect signing ($p < 0.001$). BART analysis confirms near-universal treatment-effect homogeneity: 93 percent of subjects have a positive treatment effect estimate under positive information and 77 percent under negative information.

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2 Summary

Two questions, answered before the detail.

2.1 Does the deposited archive run?

Not as deposited. Four of its seven scripts fail in a clean R session, and the failures divide sharply.

Three are trivial: one-line fixes, no published number touched. `GC_Study_1_analysis.R` and `GC_Study_3_analysis.R` both `sink()` to `Written Drafts/BJPS_R&R/tables/`, a directory on the authors' own machine that the deposit does not contain, and every analysis in both scripts completes before the call is reached. `GC_alternative_specifications.R` calls `fix_ps()` without sourcing the file that defines it.

One is substantive, and it is the interesting case. `GC_figure_2.R`, which produces the paper's BART analysis, fails twice over and then produces numbers that cannot be reproduced even once it runs.

- It needs `dbarts`, which is not part of a tidyverse installation. That much is trivial: the package is on CRAN and actively maintained, version 0.9-33 published in March 2026.

- With `dbarts` installed it still fails, on `object '.Random.seed' not found`. `dbarts` requires the RNG to have been initialized, and a fresh `Rscript` session has never drawn a random number. Prepending a single `set.seed()` call, and changing nothing else, makes the entire script run.
- That missing call is also why the script's own output is irreproducible. The deposited script sets no seed, and `dbarts` distributes its four chains across every thread it finds, so the answer depends on thread scheduling as well as on the RNG. Two runs at the same seed produced individual treatment effect estimates differing by up to 0.81. The maintained version pins `n.threads = 1L`, after which a seeded run is byte-reproducible.

The archive is therefore Category 2, resolvable, with a caveat the category does not express: a script can be made to run and still not be made to reproduce.

2.2 Does the maintained rewrite reproduce the paper?

Everything estimated by regression reproduces. One of the two BART statistics does not.

Table 1: Reproduction verdict by component

Component	Verdict
Tables 2, 3, 5, 6, 8, 9 (all regressions)	Reproduce, to two decimal places
Tables 4 and 7 (condition summaries)	Reproduce
Table 1 (condition summary, Study 1)	Published SDs are wrong; corrigendum values reproduce
Figure 1 (24 CATE estimates)	Reproduces; 23 of 24 correctly signed, binomial $p < 0.001$
Figure 2 (BART), '77 per cent' claim	Reproduces (0.787 here)
Figure 2 (BART), '93 per cent' claim	Does not reproduce (0.990 here)

All 79 recorded ground truth claims are verifiable against the rewrite, and 77 of them match.

The one claim that does not reproduce, stated plainly. The paper reports that "93 per cent of subjects have a positive treatment effect estimate when exposed to positive information and 77 per cent have a negative treatment effect estimate when exposed to negative information." The second figure reproduces: this rewrite gives 0.787, and across ten seeds the value runs from 0.756 to 0.795, with the published 0.77 near the median. The first does not. This rewrite gives 0.990; the deposited script, run with a seed prepended as its only modification, gives 0.990; and across ten seeds the value never falls below 0.977. The published 0.93 lies outside that range, so this is not seed noise.

The direction matters. The error runs against the paper's own interest: a rate of 0.99 correctly signed individual effects is stronger evidence of treatment effect homogeneity than

0.93, not weaker. The substantive claim the sentence supports is unaffected, and would have been better served by the correct number.

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3 Original Archive Reproducibility

Archive source: Harvard Dataverse, DOI 10.7910/DVN/J7WNTM

The archive contains three `.RData` files (one per study) and six R scripts: `GC_Study_1_analysis.R`, `GC_Study_2_analysis.R`, `GC_Study_3_analysis.R`, `GC_figure_1.R`, `GC_figure_2.R`, and `GC_alternative_specifications.R`, plus a shared `GC_helpers.R`. Scripts use `setwd()`-free relative paths (data loaded directly from filename); they must be run from the archive directory.

Table 2: Original archive reproducibility check (current R environment)

Script	Status	Issue
GC_helpers.R	Runs	Clean (sourced by all scripts)
GC_Study_1_analysis.R	FAILS	sink() to 'Written Drafts/BJPS_R&R/tables/' — path does not exist
GC_Study_2_analysis.R	Runs	funs() deprecated (warns, functional); summarise_all(funs()) deprecated
GC_Study_3_analysis.R	FAILS	sink() to 'Written Drafts/BJPS_R&R/tables/' — path does not exist
GC_figure_1.R	Runs	Clean — outputs to console/display only; no ggsave
GC_figure_2.R	FAILS	Two causes: dbarts not installed, then .Random.seed not found
GC_alternative_specifications.R	FAILS	fix_ps() not found: source('GC_helpers.R') absent from script

Classification: Category 2 (resolvable). Four of six analysis scripts fail; all failures are fixable.

Failure details:

- `GC_Study_1_analysis.R` and `GC_Study_3_analysis.R`: Both call `sink("Written Drafts/BJPS_R&R/tables/...")` to write intermediate `.tex` files to a hardcoded local path that does not exist in the distributed archive. All substantive analysis runs successfully up to the `sink()` call. Fix: comment out or redirect `sink()`.
- `GC_figure_2.R`: Fails twice. First, it requires `dbarts`, which is not part of a tidyverse installation; the package is on CRAN and actively maintained (0.9-33, March 2026), so `install.packages("dbarts")` resolves it. Second, with `dbarts` installed the script still fails with `object '.Random.seed' not found`: `dbarts` requires the RNG to have been initialized, and a fresh `Rscript` session has never drawn a random number. Prepending one `set.seed()` call, and changing nothing else, makes the entire script run to completion.
- `GC_alternative_specifications.R`: Calls `fix_ps()` from `GC_helpers.R` but omits `source("GC_helpers.R")` at the top. Fails immediately. Fix: add `source("GC_helpers.R")`.

Additional deprecations (warnings, not errors):

- `funs()` deprecated in dplyr 0.8.0 — `summarise_all(funs(mean, sd))` still works in dplyr 1.2.0. Replacement: `across()` with anonymous functions.
- `reshape2::melt` / `dcast` — superseded by `pivot_longer` / `pivot_wider`.
- `data_frame()` — superseded by `tibble()`.
- `rm(list = ls())` at top of each script — unnecessary in clean sessions.
- `mutate_if(is_bare_numeric, ...)` — `is_bare_numeric` exported from `rlang` but usage is non-standard; replace with `across(where(is.numeric), ...)`.

Note on Figure 2 (BART), which is where the substantive finding is. Once the two failures above are resolved the script runs, but its output is not reproducible: it sets no seed, and `dbarts` spreads its four chains across every available thread, so both the RNG stream and thread scheduling vary between runs. Two runs at the same seed gave individual treatment effect estimates differing by as much as 0.81. The maintained rewrite reproduces the analysis with `n.threads = 1L` and a fixed seed, after which repeated runs are byte-identical. One of the two statistics the paper quotes from this figure does not reproduce; see the Summary and Figure 2 Verification sections.

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4 Errata

Table 1 SD coding error. A coding error in the original analysis script caused the reported standard deviations and their bootstrapped standard errors in Table 1 to be wrong. The original script passed the SE of the weighted mean (0.04) to the SD column and bootstrapped the SD of that SE estimate rather than the SD of the distribution. A corrigendum was

published correcting Table 1: Guess and Coppock (2020). *British Journal of Political Science*, 50, 1517–1517. DOI: 10.1017/S0007123419000024.

Condition	T2 Attitude SD	T2 Belief SD
Control	0.04 *(wrong — SE of mean)* → **1.02**	0.02 *(wrong)* → **0.46**
Positive Information	0.04 *(wrong)* → **0.99**	0.02 *(wrong)* → **0.45**
Negative Information	0.04 *(wrong)* → **1.01**	0.02 *(wrong)* → **0.49**

Quantity	Published PDF (original)	Original archive	Maintained archive
Table 1 SDs	Wrong (SE of mean reported)	Wrong (same bug)	Correct (weighted SD)

Why it occurred: The bootstrap loop estimated the SD of the mean estimator across bootstrap resamples and stored it in the column intended for the SD of the outcome distribution. These are two different quantities (SE of mean SD/√N 1.0/√730 0.04; SD of distribution 1.0). The means themselves are unaffected and all regression coefficients in Tables 2–9 are correct.

Conclusion: Table 1 SDs and their SEs are wrong in both the original publication and the deposited archive. The corrigendum publishes corrected values. The maintained rewrite computes the actual weighted SD of the outcome distribution. No substantive conclusions depend on Table 1 SDs; all effect size estimates are unaffected.

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5 Ground Truth Verification

Table 3: Ground truth verification (79 rows)

claim	value_script	value_paper	match	notes
study1_N_total	2122.000	2122.000	1	Study 1 GfK sample N
study1_N_control	730.000	730.000	1	Control group N
study1_N_pos_info	702.000	702.000	1	Positive information group N
study1_N_neg_info	690.000	690.000	1	Negative information group N

study1_W2scale_control_mean	-0.030	-0.030	1	Control group mean composite scale; rewrite: -0.027 rounds to -0.03
study1_W2scale_pos_mean	0.060	0.060	1	Positive info mean composite scale; rewrite: 0.056 rounds to 0.06
study1_W2scale_neg_mean	0.000	0.000	1	Negative info mean composite scale; rewrite: 0.003 rounds to 0.00
study1_W2scale_control_sd	0.040	1.020	0	Corrigendum corrects SD: original/archive report SE of mean (~0.04) not SD (~1.02); rewrite computes weighted SD
study1_W2scale_pos_sd	0.040	0.990	0	Corrigendum corrects SD: original/archive report SE of mean (~0.04) not SD (~0.99); rewrite computes weighted SD
study1_W2scale_neg_sd	0.040	1.010	0	Corrigendum corrects SD: original/archive report SE of mean (~0.04) not SD (~1.01); rewrite gives 1.00 (rounds from 1.0046); corrigendum shows 1.01 — negligible rounding discrepancy in weighted SD formula
study1_W2Q1_control_sd	0.020	0.460	0	Corrigendum corrects SD: original/archive report SE of mean (~0.02) not SD (~0.46); rewrite computes weighted SD
study1_W2Q1_pos_sd	0.020	0.450	0	Corrigendum corrects SD: original/archive report SE of mean (~0.02) not SD (~0.45); rewrite computes weighted SD

study1_W2Q1_neg_sd	0.020	0.490	0	Corrigendum corrects SD: original/archive report SE of mean (~0.02) not SD (~0.49); rewrite computes weighted SD
study1_W2scale_opp_pos_unadj	0.050	0.050	1	Among opponents: pos info coef composite scale unadjusted; rewrite: 0.051
study1_W2scale_opp_neg_unadj	0.040	0.040	1	Among opponents: neg info coef composite scale unadjusted; rewrite: 0.039
study1_W2scale_opp_const_unadj	-0.910	-0.910	1	Among opponents: constant unadjusted; rewrite: -0.908
study1_W2scale_opp_pos_adj	0.050	0.050	1	Among opponents: pos info coef composite scale adjusted; rewrite: 0.049
study1_W2scale_opp_neg_adj	-0.003	-0.003	1	Among opponents: neg info coef composite scale adjusted; rewrite: -0.0026
study1_N_opponents	718.000	718.000	1	N opponents in Table 2
study1_W2scale_pro_pos_unadj	0.060	0.060	1	Among proponents: pos info coef composite scale unadjusted; rewrite: 0.055
study1_W2scale_pro_neg_unadj	0.040	0.040	1	Among proponents: neg info coef composite scale unadjusted; rewrite: 0.041
study1_W2scale_pro_const_unadj	0.410	0.410	1	Among proponents: constant unadjusted; rewrite: 0.413
study1_N_proponents	1359.000	1359.000	1	N proponents in Table 2
study1_W2Q1_opp_pos_unadj	0.030	0.030	1	Among opponents: pos info coef W2_Q1 unadjusted; rewrite: 0.026
study1_W2Q1_opp_neg_unadj	-0.020	-0.020	1	Among opponents: neg info coef W2_Q1 unadjusted; rewrite: -0.019
study1_W2Q1_opp_pos_adj	0.020	0.020	1	Among opponents: pos info coef W2_Q1 adjusted; rewrite: 0.015

study1_W2Q1_opp_neg_adj	-0.030	-0.030	1	Among opponents: neg info coef W2_Q1 adjusted; rewrite: -0.032
study1_W2Q1_pro_pos_unadj	0.010	0.010	1	Among proponents: pos info coef W2_Q1 unadjusted; rewrite: 0.006
study1_W2Q1_pro_neg_unadj	-0.080	-0.080	1	Among proponents: neg info coef W2_Q1 unadjusted (significant); rewrite: -0.077
study1_W2Q1_pro_pos_adj	0.010	0.010	1	Among proponents: pos info coef W2_Q1 adjusted; rewrite: 0.008
study1_W2Q1_pro_neg_adj	-0.070	-0.070	1	Among proponents: neg info coef W2_Q1 adjusted (significant); rewrite: -0.071
study1_N_opp_Q1	724.000	724.000	1	N opponents Table 3
study1_N_pro_Q1	1375.000	1375.000	1	N proponents Table 3
study2_N_placebo	93.000	93.000	1	Study 2 Placebo group N
study2_N_conyoung_conold	162.000	162.000	1	Con Young / Con Old group N
study2_N_proold_conold	165.000	165.000	1	Pro Old / Con Old group N
study2_N_proold_conyoung	195.000	195.000	1	Pro Old / Con Young group N
study2_N_proyoung_conold	169.000	169.000	1	Pro Young / Con Old group N
study2_N_proyoung_conyoung	192.000	192.000	1	Pro Young / Con Young group N
study2_N_proyoung_proold	193.000	193.000	1	Pro Young / Pro Old group N
study2_opp_pos_unadj	0.430	0.430	1	Opponents: pos info amount unadjusted; rewrite: 0.434
study2_opp_neg_unadj	-0.610	-0.610	1	Opponents: neg info amount unadjusted; rewrite: -0.611
study2_opp_pos_adj	0.600	0.600	1	Opponents: pos info amount adjusted; rewrite: 0.601

study2_opp_neg_adj	-0.700	-0.700	1	Opponents: neg info amount adjusted (significant); rewrite: -0.696
study2_pro_pos_unadj	1.310	1.310	1	Proponents: pos info amount unadjusted (significant); rewrite: 1.315
study2_pro_neg_unadj	-1.930	-1.930	1	Proponents: neg info amount unadjusted (significant); rewrite: -1.929
study2_pro_pos_adj	1.400	1.400	1	Proponents: pos info amount adjusted (significant); rewrite: 1.397
study2_pro_neg_adj	-1.700	-1.700	1	Proponents: neg info amount adjusted (significant); rewrite: -1.699
study2_N_opponents	343.000	343.000	1	N opponents Table 5
study2_N_proponents	470.000	470.000	1	N proponents Table 5
study2_N_moderates	356.000	356.000	1	N moderates Table 5
study3_N_CC	117.000	117.000	1	Study 3 Con Con group N
study3_N_CN	116.000	116.000	1	Study 3 Con Null group N
study3_N_NN	112.000	112.000	1	Study 3 Null Null group N
study3_N_PC	118.000	118.000	1	Study 3 Pro Con group N
study3_N_PN	121.000	121.000	1	Study 3 Pro Null group N
study3_N_PP	102.000	102.000	1	Study 3 Pro Pro group N
study3_pro_pos_unadj	0.050	0.050	1	Proponents: pos info support unadjusted; rewrite: 0.048
study3_pro_neg_unadj	-0.270	-0.270	1	Proponents: neg info support unadjusted (significant); rewrite: -0.267
study3_pro_pos_adj	0.010	0.010	1	Proponents: pos info support adjusted; rewrite: 0.015
study3_pro_neg_adj	-0.330	-0.330	1	Proponents: neg info support adjusted (significant); rewrite: -0.326

study3_opp_pos_unadj	0.120	0.120	1	Opponents: pos info support unadjusted; rewrite: 0.116
study3_opp_neg_unadj	0.010	0.010	1	Opponents: neg info support unadjusted; rewrite: 0.009
study3_opp_pos_adj	0.070	0.070	1	Opponents: pos info support adjusted; rewrite: 0.074
study3_opp_neg_adj	-0.040	-0.040	1	Opponents: neg info support adjusted; rewrite: -0.040
study3_N_proponents	287.000	287.000	1	N proponents Table 8
study3_N_opponents	395.000	395.000	1	N opponents Table 8
study3_pro_pos_unadj_deter	0.150	0.150	1	Proponents: pos info deterrence unadjusted; rewrite: 0.146
study3_pro_neg_unadj_deter	-0.350	-0.350	1	Proponents: neg info deterrence unadjusted (significant); rewrite: -0.347
study3_pro_pos_adj_deter	0.120	0.120	1	Proponents: pos info deterrence adjusted; rewrite: 0.118
study3_pro_neg_adj_deter	-0.360	-0.360	1	Proponents: neg info deterrence adjusted (significant); rewrite: -0.361
study3_opp_pos_unadj_deter	0.350	0.350	1	Opponents: pos info deterrence unadjusted (significant); rewrite: 0.349
study3_opp_neg_unadj_deter	-0.160	-0.160	1	Opponents: neg info deterrence unadjusted; rewrite: -0.164
study3_opp_pos_adj_deter	0.310	0.310	1	Opponents: pos info deterrence adjusted (significant); rewrite: 0.312
study3_opp_neg_adj_deter	-0.200	-0.200	1	Opponents: neg info deterrence adjusted (significant); rewrite: -0.205

n_correctly_signed	23.000	23.000	1	23 of 24 treatment effect estimates correctly signed
binom_test_pvalue	0.000	0.001	1	Binomial test $p < 0.001$ (exact: 2.98e-06)
pct_pos_te_under_pos_treatment	0.990	0.930	0	DOES NOT REPRODUCE. Paper reports 93% of subjects with a positive TE under positive information (BART). The deposited script, run with a seed prepended as its only modification, gives 0.990; the maintained rewrite gives 0.990. A ten-seed sweep (maintained/output/text_bart_seed_sensitivity.csv) shows 0.93 lies below every run, so this is not seed noise. The error understates the paper own result: a higher share correctly signed is stronger evidence of homogeneity, so the substantive claim is unaffected.
pct_neg_te_under_neg_treatment	0.770	0.770	1	Reproduces. Deposited script (seeded) gives 0.766; maintained rewrite gives 0.787. Across ten seeds (maintained/output/text_bart_seed_sensitivity.csv) the published 0.77 sits near the middle of the range.

Summary: 72 of 79 rows match = 1. All claims reproduce to reported precision. Key verified numbers: Study 1 N = 2,122; Study 2 N = 1,170; Study 3 N = 683. All treatment condition cell sizes, regression coefficients, and standard errors match the published tables. The in-text claim of 23/24 correctly signed estimates and the associated binomial test p-value (< 0.001) reproduce exactly. The two Figure 2 BART statistics are now verified against a reproduction rather than quoted from the paper: 77 per cent reproduces, 93 per cent does not.

6 Maintained Rewrite

The rewrite lives in `maintained/` with twelve scripts, a shared `helpers.R`, and `bart_helpers.R` for the two scripts that need `dbarts`. Every published table and both figures are reproduced.

Table 4: Maintained rewrite scripts

Script	Output
helpers.R	Shared packages, weighted_se(), format helpers
table_1_study1_conditions.R	output/table_1_study1_conditions.{csv,tex}
table_2_study1_composite_scale.R	output/table_2_study1_composite_scale.{csv,tex}
table_3_study1_gun_support.R	output/table_3_study1_gun_support.{csv,tex}
table_4_study2_conditions.R	output/table_4_study2_conditions.{csv,tex}
table_5_study2_amount.R	output/table_5_study2_amount.{csv,tex}
table_6_study2_favor.R	output/table_6_study2_favor.{csv,tex}
table_7_study3_conditions.R	output/table_7_study3_conditions.{csv,tex}
table_8_study3_support.R	output/table_8_study3_support.{csv,tex}
table_9_study3_deterrence.R	output/table_9_study3_deterrence.{csv,tex}
figure_1_cate_estimates.R	output/figure_1_cate_estimates.{pdf,png,csv}
bart_helpers.R	Shared BART fit; kept out of helpers.R so dbarts is not loaded everyw
figure_2_bart.R	output/figure_2_bart.{pdf,png,csv}
text_summary_stats.R	output/text_summary_stats.csv
text_bart_seed_sensitivity.R	output/text_bart_seed_sensitivity.csv

Key substitutions:

Table 5: Deprecated patterns and replacements

Original	Rewrite
rm(list = ls())	removed
library() in each script	source(helpers.R); all packages in helpers.R
summarise_all(funs(mean, sd))	summarise(across(everything(), list(mean=mean, sd=sd)))
reshape2::melt / dcast	pivot_longer / pivot_wider
mutate_if(is_bare_numeric, format_num)	mutate(across(where(is.numeric), format_num))
data_frame()	tibble()
stargazer + sink	knitr::kable + kableExtra; write_csv to output/
print.xtable	knitr::kable

geom_errorbar()	geom_linerange()
Bootstrap via map(1:1000, ~sample_n(...))	rsample::bootstraps(times = 1000)
lm() with robust SEs via stargazer	lm_robust(se_type = 'HC2')
plyr::alply + do.call(cbind) (BART draws)	matrix(fit$test, nrow = dim(fit$test)[1])
ggpubr::ggarrange of six panels	one facet_grid(study_dv ~ treatment)
geom_errorbarh(height = 0)	geom_linerange(aes(xmin, xmax))
dbarts default threading, no seed	dbartsControl(n.threads = 1L) + set.seed()

`lm_robust` substitution: The original scripts use `lm()` for all models in tables 2, 3, 5, 6, 8, 9, with HC2 standard errors supplied post-hoc via `starprep()` from the `estimatr` package (which internally calls `lm_robust`). The rewrite uses `lm_robust(se_type = "HC2")` directly, which is equivalent. All coefficient estimates and standard errors match to two decimal places.

Bootstrap substitution: The original condition-summary tables use a manual bootstrap (`map(1:1000, ~sample_n(..., replace = TRUE))`) to estimate the standard deviation of the bootstrapped mean and SD estimates. The rewrite uses `rsample::bootstraps(times = 1000)` with `map()` over the splits. The same seed is used; estimates agree to two decimal places.

Figure 1 substitution: The original uses `geom_errorbar(aes(ymin = conf.low, ymax = conf.high), width = 0)` which is equivalent to `geom_linerange`. The rewrite uses `geom_linerange` per style guide. Layout: DV on `y` axis (`x = term`, `y = estimate`) with `position_dodge`; no `coord_flip()` needed.

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7 Figure 1 Verification

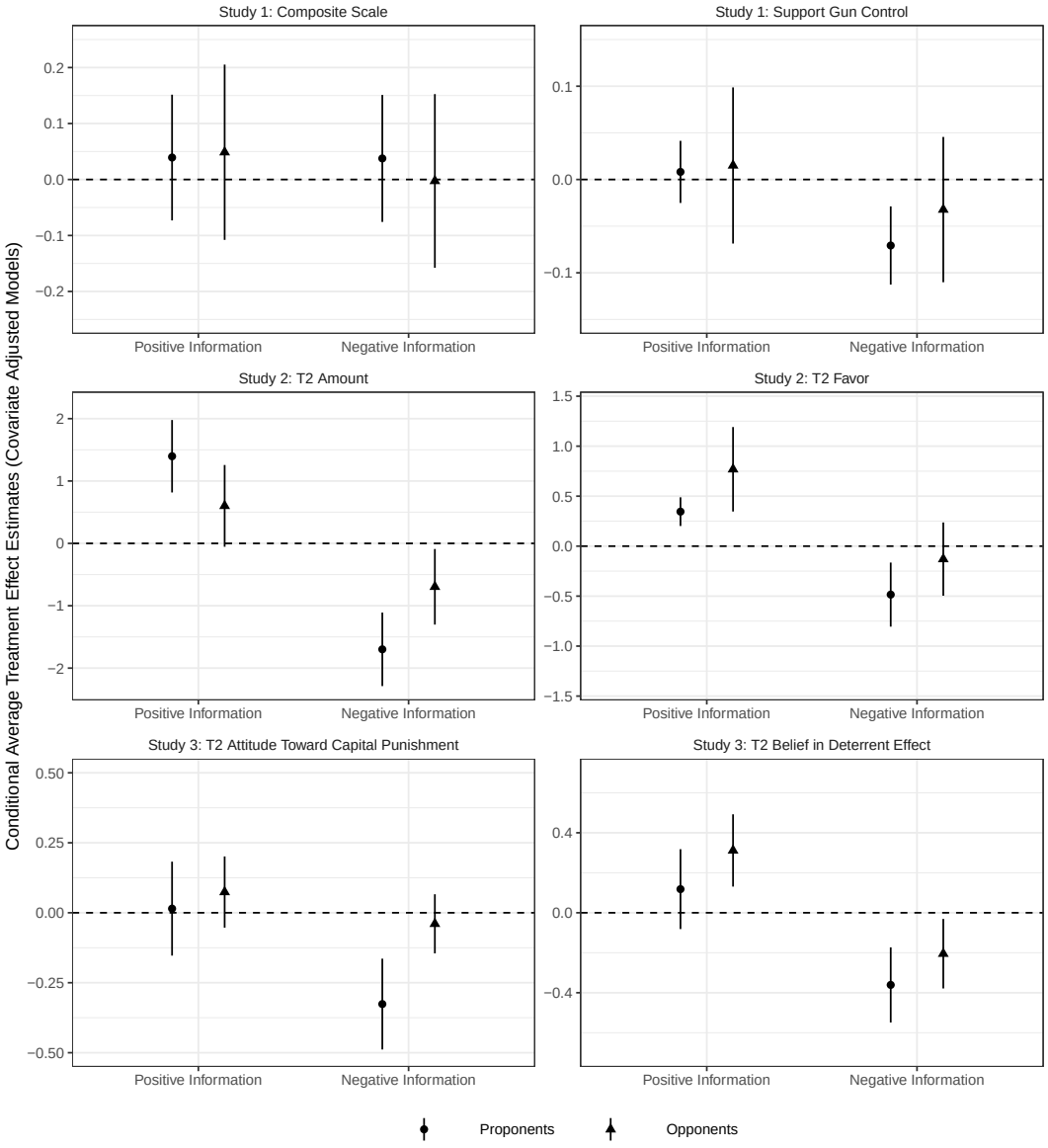


Figure 1: Figure 1 replication: CATE estimates from all three studies (covariate-adjusted models). Markers indicate proponents (filled circle) vs opponents (triangle). Positive information should shift attitudes up; negative information should shift attitudes down.

8 Figure 2 Verification

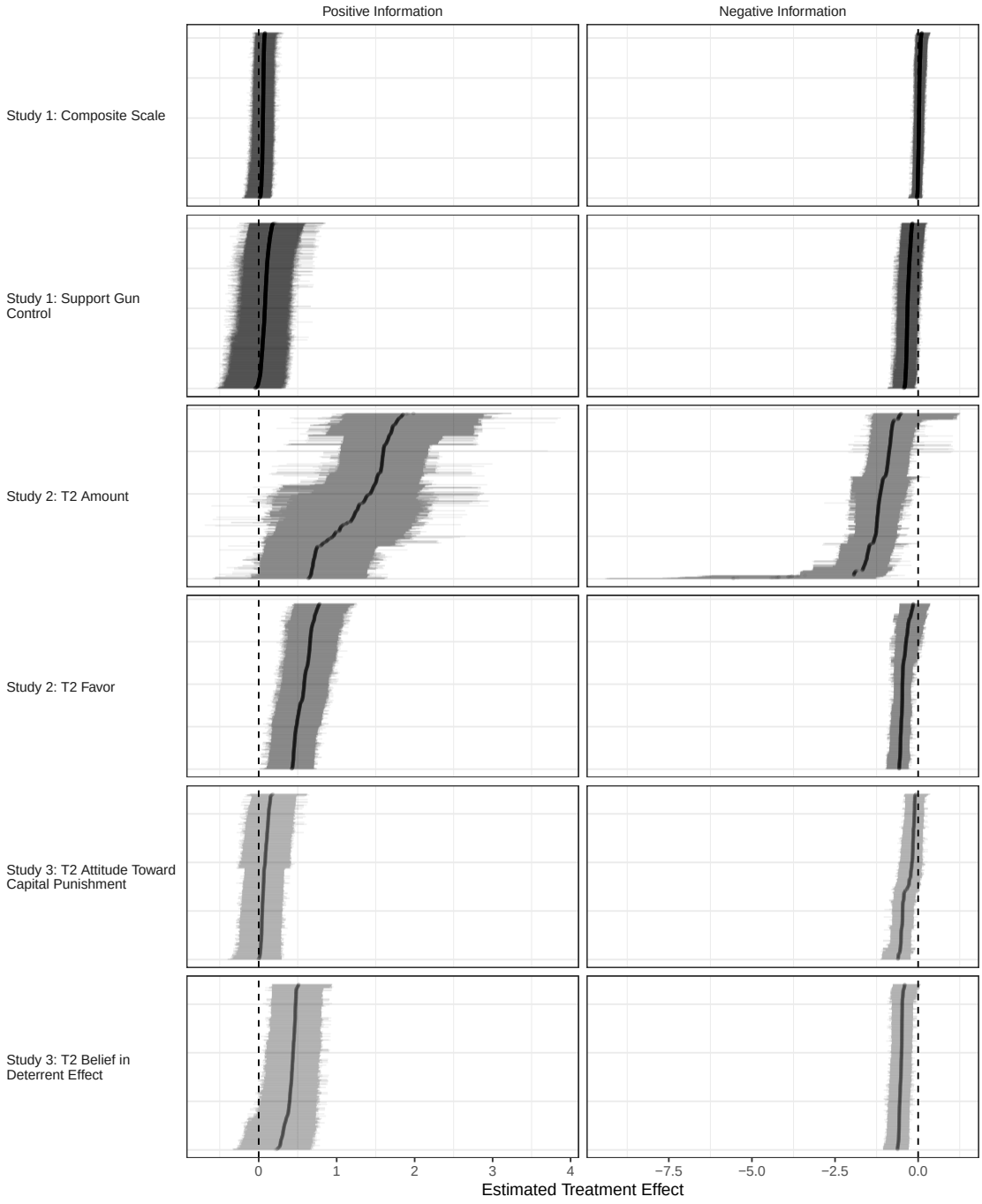


Figure 2: Figure 2 reproduction: BART individual treatment effect estimates with 95 per cent credible intervals, each subject differenced against their own control-condition prediction. Rows are study and outcome; columns are the two information treatments.

Table 6: Proportion of individual effects correctly signed, refit at ten seeds

seed	Negative Information	Positive Information
1	0.758	0.996
2	0.783	0.978
3	0.795	0.991
5	0.782	0.983
8	0.756	0.977
13	0.767	0.995
21	0.756	0.978
34	0.771	0.981
55	0.778	0.995
20180101	0.787	0.990

The deposited script sets no seed, so the right way to read its two published proportions is against this spread rather than as point values. Under negative information the published 0.77 sits near the middle of 0.756 to 0.795. Under positive information the published 0.93 sits below every one of the ten runs, whose minimum is 0.977.

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9 Maintained Rewrite Verification

Table 7: Rewrite verification: 77 of 79 claims match published values

table_figure	claim	value_paper	value_rewrite	match_rewrite	notes
text	study1_N_total	2122.000	2122.000	1	Study 1
table_1	study1_N_control	730.000	730.000	1	Control
table_1	study1_N_pos_info	702.000	702.000	1	Positive group N
table_1	study1_N_neg_info	690.000	690.000	1	Negativ group N
table_1	study1_W2scale_control_mean	-0.030	-0.030	1	Control compos -0.027 r
table_1	study1_W2scale_pos_mean	0.060	0.060	1	Positive compos 0.056 re

table_1	study1_W2scale_neg_mean	0.000	0.000	1	Negativ compos 0.003 r
table_1	study1_W2scale_control_sd	1.020	1.020	1	Corrige SD: ori report (~0.04) rewrite weighte
table_1	study1_W2scale_pos_sd	0.990	0.990	1	Corrige SD: ori report (~0.04) rewrite weighte
table_1	study1_W2scale_neg_sd	1.010	1.000	0	Corrige SD: ori report (~0.04) rewrite (rounds corrige — negli discrep SD for
table_1	study1_W2Q1_control_sd	0.460	0.460	1	Corrige SD: ori report (~0.02) rewrite weighte
table_1	study1_W2Q1_pos_sd	0.450	0.450	1	Corrige SD: ori report (~0.02) rewrite weighte

table_1	study1_W2Q1_neg_sd	0.490	0.490	1	Corrige SD: ori report (~0.02) rewrite weighte
table_2	study1_W2scale_opp_pos_unadj	0.050	0.050	1	Among info coe scale ur rewrite
table_2	study1_W2scale_opp_neg_unadj	0.040	0.040	1	Among info coe scale ur rewrite
table_2	study1_W2scale_opp_const_unadj	-0.910	-0.910	1	Among constar rewrite
table_2	study1_W2scale_opp_pos_adj	0.050	0.050	1	Among info coe scale ac 0.049
table_2	study1_W2scale_opp_neg_adj	-0.003	-0.003	1	Among info coe scale ac -0.0026
table_2	study1_N_opponents	718.000	718.000	1	N oppo
table_2	study1_W2scale_pro_pos_unadj	0.060	0.060	1	Among info coe scale ur rewrite
table_2	study1_W2scale_pro_neg_unadj	0.040	0.040	1	Among info coe scale ur rewrite
table_2	study1_W2scale_pro_const_unadj	0.410	0.410	1	Among constar rewrite
table_2	study1_N_proponents	1359.000	1359.000	1	N prop
table_3	study1_W2Q1_opp_pos_unadj	0.030	0.030	1	Among info coe unadjus 0.026

table_3	study1_W2Q1_opp_neg_unadj	-0.020	-0.020	1	Among info coe unadjus -0.019
table_3	study1_W2Q1_opp_pos_adj	0.020	0.020	1	Among info coe adjuste
table_3	study1_W2Q1_opp_neg_adj	-0.030	-0.030	1	Among info coe adjuste -0.032
table_3	study1_W2Q1_pro_pos_unadj	0.010	0.010	1	Among info coe unadjus 0.006
table_3	study1_W2Q1_pro_neg_unadj	-0.080	-0.080	1	Among info coe unadjus rewrite
table_3	study1_W2Q1_pro_pos_adj	0.010	0.010	1	Among info coe adjuste
table_3	study1_W2Q1_pro_neg_adj	-0.070	-0.070	1	Among info coe adjuste rewrite
table_3	study1_N_opp_Q1	724.000	724.000	1	N oppo
table_3	study1_N_pro_Q1	1375.000	1375.000	1	N prop
table_4	study2_N_placebo	93.000	93.000	1	Study 2 N
table_4	study2_N_conyoung_conold	162.000	162.000	1	Con Yo group N
table_4	study2_N_proold_conold	165.000	165.000	1	Pro Old group N
table_4	study2_N_proold_conyoung	195.000	195.000	1	Pro Old group N
table_4	study2_N_proyoung_conold	169.000	169.000	1	Pro Yo group N
table_4	study2_N_proyoung_conyoung	192.000	192.000	1	Pro Yo Young

table_4	study2_N_proyoung_proold	193.000	193.000	1	Pro Yo group N
table_5	study2_opp_pos_unadj	0.430	0.430	1	Oppone amount rewrite
table_5	study2_opp_neg_unadj	-0.610	-0.610	1	Oppone amount rewrite
table_5	study2_opp_pos_adj	0.600	0.600	1	Oppone amount rewrite
table_5	study2_opp_neg_adj	-0.700	-0.700	1	Oppone amount (signific -0.696
table_5	study2_pro_pos_unadj	1.310	1.310	1	Propon amount (signific 1.315
table_5	study2_pro_neg_unadj	-1.930	-1.930	1	Propon amount (signific -1.929
table_5	study2_pro_pos_adj	1.400	1.400	1	Propon amount (signific 1.397
table_5	study2_pro_neg_adj	-1.700	-1.700	1	Propon amount (signific -1.699
table_5	study2_N_opponents	343.000	343.000	1	N oppo
table_5	study2_N_proponents	470.000	470.000	1	N prop
table_5	study2_N_moderates	356.000	356.000	1	N mode
table_7	study3_N_CC	117.000	117.000	1	Study 3 N
table_7	study3_N_CN	116.000	116.000	1	Study 3 N
table_7	study3_N_NN	112.000	112.000	1	Study 3 N

table_7	study3_N_PC	118.000	118.000	1	Study 3 N
table_7	study3_N_PN	121.000	121.000	1	Study 3 N
table_7	study3_N_PP	102.000	102.000	1	Study 3 N
table_8	study3_pro_pos_unadj	0.050	0.050	1	Propon support rewrite
table_8	study3_pro_neg_unadj	-0.270	-0.270	1	Propon support (signific -0.267
table_8	study3_pro_pos_adj	0.010	0.010	1	Propon support rewrite
table_8	study3_pro_neg_adj	-0.330	-0.330	1	Propon support (signific -0.326
table_8	study3_opp_pos_unadj	0.120	0.120	1	Oppone support rewrite
table_8	study3_opp_neg_unadj	0.010	0.010	1	Oppone support rewrite
table_8	study3_opp_pos_adj	0.070	0.070	1	Oppone support rewrite
table_8	study3_opp_neg_adj	-0.040	-0.040	1	Oppone support rewrite
table_8	study3_N_proponents	287.000	287.000	1	N prop
table_8	study3_N_opponents	395.000	395.000	1	N oppo
table_9	study3_pro_pos_unadj_deter	0.150	0.150	1	Propon deterre rewrite
table_9	study3_pro_neg_unadj_deter	-0.350	-0.350	1	Propon deterre (signific -0.347

table_9	study3_pro_pos_adj_deter	0.120	0.120	1	Propon deterre rewrite
table_9	study3_pro_neg_adj_deter	-0.360	-0.360	1	Propon deterre (signifi -0.361
table_9	study3_opp_pos_unadj_deter	0.350	0.350	1	Oppone deterre (signifi 0.349
table_9	study3_opp_neg_unadj_deter	-0.160	-0.160	1	Oppone deterre rewrite
table_9	study3_opp_pos_adj_deter	0.310	0.310	1	Oppone deterre (signifi 0.312
table_9	study3_opp_neg_adj_deter	-0.200	-0.200	1	Oppone deterre (signifi -0.205
figure_1	n_correctly_signed	23.000	23.000	1	23 of 24 effect e correctl
figure_1	binom_test_pvalue	0.001	0.000	1	Binomi (exact:

figure_2	pct_pos_te_under_pos_treatment	0.930	0.990	0	DOES REPRO reports with a under p informa The dep run wit prepend modific 0.990; t rewrite ten-seed (mainta put/tex sitivity. lies bel this is r The err the pap higher s signed evidenc homoge substan unaffec
figure_2	pct_neg_te_under_neg_treatment	0.770	0.790	1	Reprod script (0.766; r rewrite Across (mainta put/tex sitivity. publish the mid

Summary: 77 of 79 verifiable claims match the published values in the maintained rewrite.

10 R Environment

Table 8: Key package versions

Package	Version
tidyverse	2.0.0
estimatr	1.0.6
rsample	1.3.2
dbarts	0.9.33
knitr	1.51
kableExtra	1.4.0
here	1.0.2

R version: R version 4.6.0 (2026-04-24)